

02_Workflow_v1.5.md

MonitorAI v1.5 Workflow Specification

Version: 1.5

Status: Baseline

1. Workflow Overview

MonitorAI uses a standardized image acquisition workflow.

All patients follow the same sequence.

The workflow is designed to maximize:

- Standardization
- Repeatability
- Dataset Quality

while minimizing patient burden.

2. Workflow Sequence

Launch App

↓

Authentication

↓

Patient Selection

↓

Position Calibration

↓

11-Step Capture Workflow



Review Screen



Upload



Completion Screen

3. Position Calibration

Position Calibration is mandatory.

Purpose:

Standardize initial positioning before image acquisition.

Display Elements

- Live Camera Preview
 - 3 × 3 Grid
 - Center Marker
 - Opening Detection Overlay
-

Instruction

撮影口を画面中央に合わせてください

Success Criteria

Opening Center Error < 5%

AND

Stable for 3 Seconds



Automatic Advance

Metadata

Store:

- Calibration Start Time
 - Calibration End Time
 - Calibration Duration
 - Opening Offset X
 - Opening Offset Y
 - Calibration Quality Score
-

4. Capture Workflow

The workflow contains 11 capture steps.

Group A

Aligner ON

5 mm Open

Step01

Front ON

Objective:

Capture frontal aligner fit.

Overlay:

Front_ON.png

Step02

Right Buccal ON

Objective:

Capture right posterior aligner fit.

Overlay:

Right_Buccal_ON.png

Step03

Left Buccal ON

Objective:

Capture left posterior aligner fit.

Overlay:

Left_Buccal_ON.png

Group B

Aligner OFF

5 mm Open

Step04

Front OFF

Overlay:

Front_OFF.png

Step05

Right Buccal OFF

Overlay:

Right_Buccal_OFF.png

Step06

Left Buccal OFF

Overlay:

Left_Buccal_OFF.png

Step07

Upper Occlusal

Overlay:

Upper_Occlusal.png

Objective:

Maximum upper arch visibility.

Step08

Lower Occlusal

Overlay:

Lower_Occlusal.png

Objective:

Maximum lower arch visibility.

Group C

Occlusion

Light Bite

Step09

Front Occlusion

Overlay:

Front_Occlusion.png

Step10

Right Occlusion

Overlay:

Right_Occlusion.png

Step11

Left Occlusion

Overlay:

Left_Occlusion.png

5. PTG Workflow

Every step uses PTG.

PTG evaluates:

Clinical Visibility

rather than

Head Rotation Angle.

PTG Levels

Dot1

Poor



Dot2

Improving



Dot3

Acceptable



Dot4

Minimum Passing



Dot5

Recommended



Dot6

Ideal

6. Visibility Evaluation

Buccal Steps

Evaluate:

- Posterior Tooth Visibility
- Molar Visibility
- Clinical Usefulness

Used For:

Step02

Step03

Step05

Step06

Step10

Step11

Occlusal Steps

Evaluate:

- Arch Visibility
- Posterior Arch Visibility
- Occlusal Surface Visibility

Used For:

Step07

Step08

7. Auto Capture

Manual shutter operation is discouraged.

The system performs automatic capture.

Capture Rule

PTG $\geq$ Dot4

AND

Stable 4 Seconds

↓

Capture

Validation Before Capture

Verify:

- Focus
 - Blur
 - Visibility
 - Exposure
-

8. Plateau Detection

Plateau Detection is mandatory.

Purpose:

Avoid excessive retakes.

Plateau Rule

PTG $\geq$ Dot4

AND

No Visibility Improvement

AND

Stable 4 Seconds

↓

Plateau Capture

Voice Message

十分良い位置です

この位置で撮影します

9. Review Workflow

After Step11:

Display Review Screen.

Display

Show:

- 11 Images
 - Capture Status
 - Retake Controls
-

Retake Policy

Patients may retake individual images.

Entire session retake is not required.

10. Upload Workflow

Upload begins only after patient confirmation.

Upload Sequence

Authenticate

↓

Create Session

↓

Create Capture Record

↓

Upload Image

↓

Upload Metadata

↓

Repeat

↓

Complete Session

↓

Logout

11. Offline Workflow

Network availability is not required during capture.

Offline Process

Capture

↓

Store Locally

↓

Queue Upload

↓

Reconnect

↓

Automatic Upload

Retry Schedule

1 Minute



5 Minutes



15 Minutes



1 Hour



Manual Retry

12. Completion Workflow

When upload succeeds:

Display Completion Screen.

Message

撮影が完了しました

ご協力ありがとうございました

13. Workflow Success Criteria

A workflow is successful when:

- Position Calibration Complete

- 11 Images Captured
- Metadata Recorded
- Upload Successful

and

No Data Loss Occurs.

14. Final Workflow Principle

The objective is not:

Taking photographs.

The objective is:

Creating standardized orthodontic clinical records suitable for future AI dataset generation.